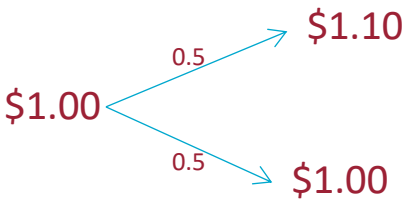
Financial Models: Dealing with Risk & Uncertainty

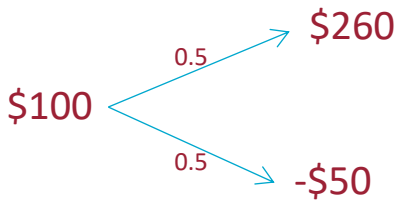
© Mohammad M. Rahaman, PhD
Canada Research Chair (CRC) in International Finance & Competitiveness
Associate Dean, Strategic Partnership & Community Engagement
Professor of Finance
Sobey School of Business
Saint Mary's University

Sobey School of Business
Saint Mary's University

Why Risk Analyses in Finance?

- You are given the following two gambles. Which one would you pick and why?



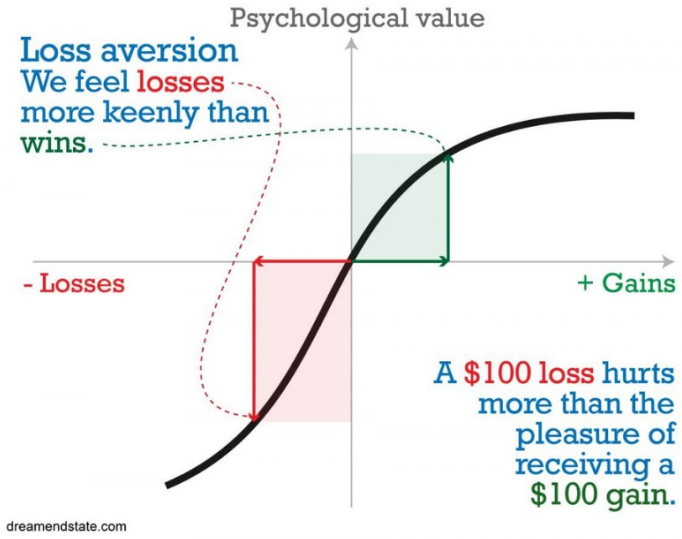


© M.M. Rahaman

The Prospect Theory

The prospect theory suggests that people feel losses twice as hard as gains.

Many people don't want to play a Heads or Tails game where they can win $100 but risk losing $50.



You should take this bet every single day.

Why Risk Analyses in Finance?

- When you deal with uncertainty then a decision criterion based on a ***SINGLE*** number is no good.
  - Go ahead if NPV>0; Reject if NOV<0.
  - Go ahead if IRR>Hurdle rate; Reject if IRR<Hurdle rate.

- Monte Carlo Simulation allows you to generate the whole distribution of values (in the case of the example, return), rather than only one number.

© M.M. Rahaman

Probabilistic Approaches in Financial Modelling

- In reality, the outcomes from an investment (based on a valuation model) is always uncertain and this is the he essence of RISK in finance. There two ways of dealing with risk in financial modelling:
  - **The risk adjusted cash flow approach:**
    - We make the adjustment by either raising discount rates or lowering cash flows.
  - **Probabilistic approaches:**
    - We deal with uncertainty more explicitly by asking what if questions about key inputs and looking at the impact on value (Sensitivity Analysis). We looking at the cash flows/value under different scenarios for the future (Scenario Analysis). We use probability distributions for key inputs, rather than expected values, and computing value as a distribution as well (Simulations).

© M.M. Rahaman

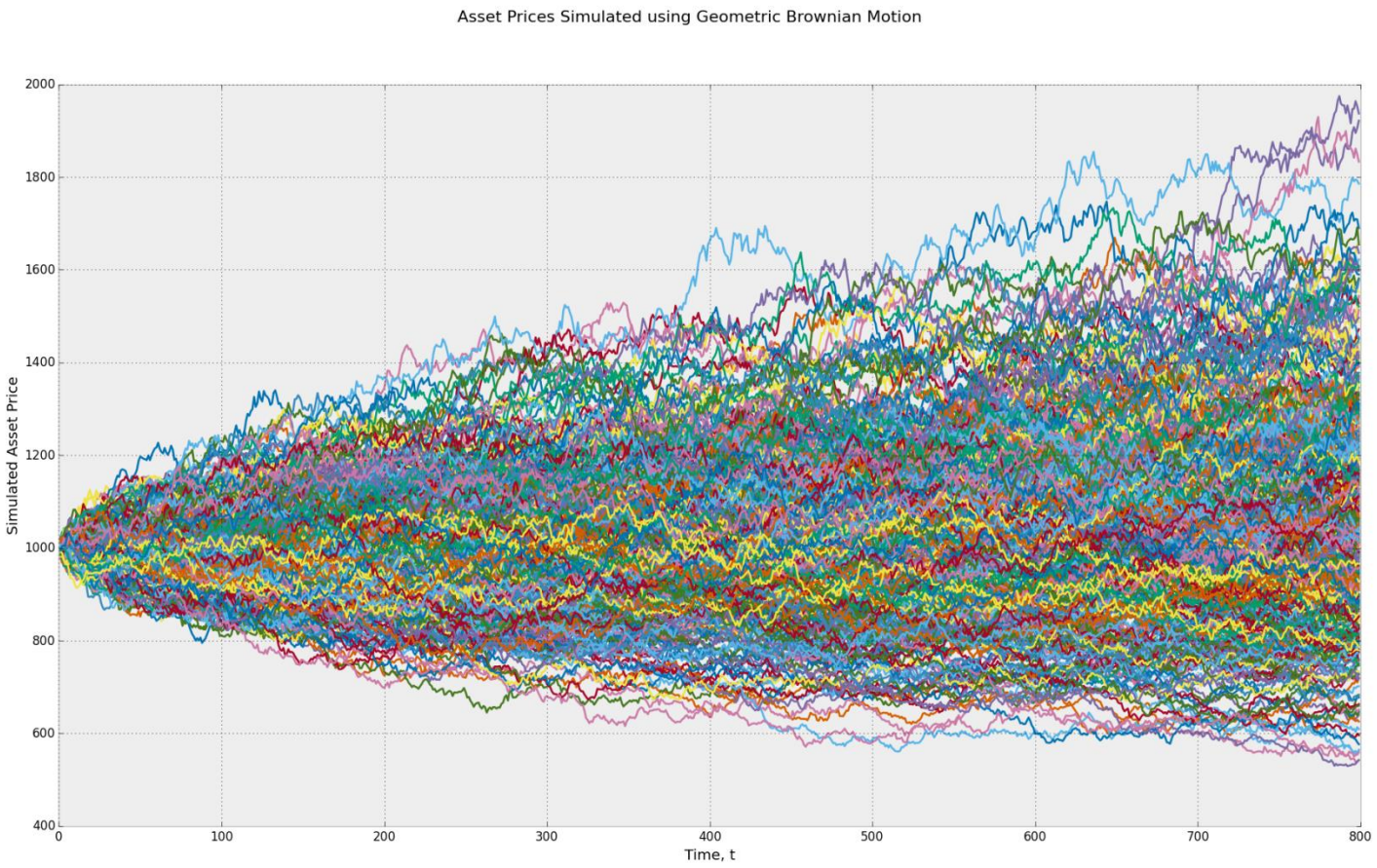
Dealing with Risk and Uncertainty

RISK versus UNCERTAINTY
Are they the same or different?

$$dS_t = \overbrace{S_t\,\mu\,dt}^{\text{drift}} + \underbrace{S_t\,\sigma\,\varepsilon\sqrt{\Delta t}}_{\text{uncertainty}}$$

© M.M. Rahaman

Dealing with Risk and Uncertainty



© M.M. Rahaman

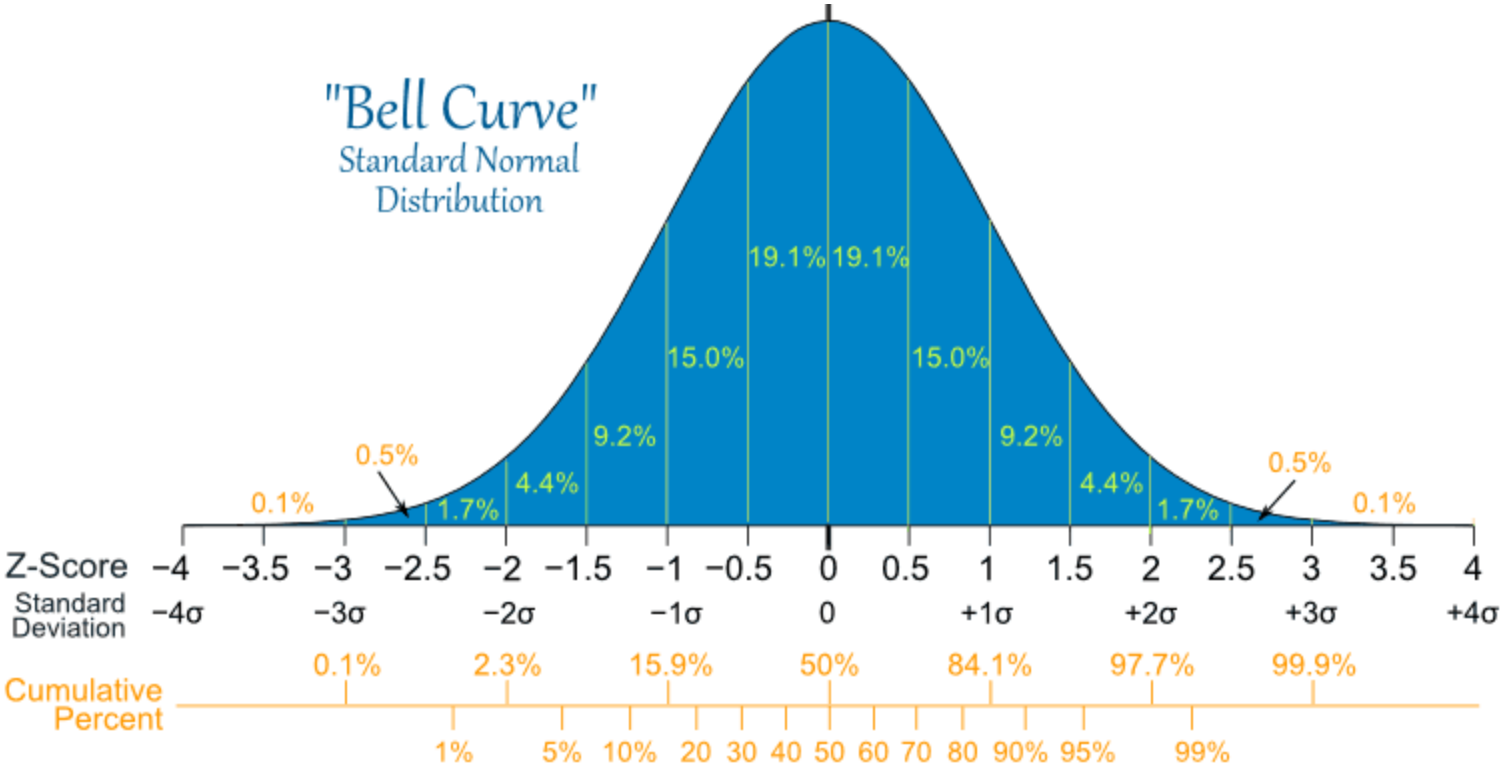
Dealing with Risk and Uncertainty

RISK versus UNCERTAINTY
Are they the same or different?

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(\mu - x)^2}{2\sigma^2}}$$

© M.M. Rahaman

Dealing with Risk and Uncertainty



© M.M. Rahaman

Forecasting Asset Prices

- Log-Normal model forecast
- Bootstrap forecast
- Mean-reverting forecast
- Moving average forecast
- Regression-based forecast

© M.M. Rahaman

Log-Normal Model of Stock Return Forecast

$$dS_t = \overbrace{S_t\,\mu\,dt}^{\text{drift}} + \underbrace{S_t\,\sigma\,\varepsilon\sqrt{\Delta t}}_{\text{uncertainty}}$$

- In this model:
  - S_t=Current stock price
  - $\mu S_t \Delta t$=Mean evolution in stock price
  - $\sigma S_t \sqrt{\Delta t}$ =Standard deviation of future movement in stock price
  - ε is a random variable that is normally distributed with mean 0 and standard deviation 1, i.e., the white noise.
  - Note that when Δt is really small, standard deviation of the stock's movement will greatly exceed its mean because for really small Δt, $\sqrt{\Delta t}$ will be much larger than Δt.

© M.M. Rahaman

Log-Normal Model of Stock Return Forecast

- Again, recall from your earlier classes, when Δt is really small, the natural logarithm ($Ln(S_t)$) of stock price will **change** (by Ito's Lemma) by an amount that is **normally distributed** with:
  - Mean = $\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t$ and
  - Standard deviation = $\sigma\sqrt{\Delta t}$
  - Where $\left(\mu - \frac{1}{2}\sigma^2\right)$ is the continuously compounded rate of return on the stock. Also note that the continuously compounded rate of return on the stock is less than the instantaneous return.
- But we want to forecast the actual stock price. It can be shown that for a time interval t, the logarithm of stock price, $Ln(S_t)$, follows a normal distribution with:
  - Mean = $Ln(S_0) + \left(\mu - \frac{1}{2}\sigma^2\right)t$ and
  - Standard deviation = $\sigma\sqrt{t}$

© M.M. Rahaman

Log-Normal Model of Stock Return Forecast

- If $Ln(S_t)$, follows a normal distribution with:
  - Mean = $Ln(S_0) + \left(\mu - \frac{1}{2}\sigma^2\right)t$ and
  - Standard deviation = $\sigma\sqrt{t}$
- S_t will follow a **lognormal distribution**. In other words, our future stock price will be a lognormal **random variable**.
- The future evolution in $Ln(S_t)$ is given by:
  - $Ln(S_t) = Ln(S_0) + \left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma^2\sqrt{t}\,\varepsilon$
- Therefore, to get to S_t we must take the antilog so that:
  - $S_t = S_0 e^{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma\sqrt{t}\,\epsilon}$

© M.M. Rahaman

Log-Normal Model of Stock Return Forecast

- Since we know that future path of our log-normally distributed stock price, how do we actually go about actually forecasting the future stock price?
- It turns out that the crucial parameters in forecasting future stock price using the log-normal model are: $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$.
- If we know these two parameters we can generate the future path of our stock prices.

© M.M. Rahaman

Log-Normal Model of Stock Return Forecast

- How do you estimate $\boldsymbol{\mu}$?
  - It is extremely difficult to estimate the instantaneous drift. If you know $\boldsymbol{\mu}$, then you essentially know the future mean return of the stock.
  - The best we can do is to assume some equally likely value for $\boldsymbol{\mu}$.

© M.M. Rahaman

Log-Normal Model of Stock Return Forecast

- How do you estimate $\boldsymbol{\sigma}$?
  - We can actually do a much better job in estimating $\boldsymbol{\sigma}$ than estimating $\boldsymbol{\mu}$ accurately.
  - We can simply find the $\boldsymbol{\sigma}$ from the implied volatility of the put or call option traded in the market.
  - Simply making sure that the Black-Scholes option price of the put or call match the market price of the option, we can figure out the implied volatility of the stock price and use that as an estimate of $\boldsymbol{\sigma}$ in our forecasting model.

$$Call\ Premium = S \times N(d_1) - K \times exp^{-r(T-t)} \times N(d_2)$$
$$Put\ Premium = K \times exp^{-r(T-t)} \times N(-d_2) - S \times N(-d_1)$$
$$S = Spot\ price\ of\ the\ underlying\ asset$$
$$K = Strike\ price$$
$$r = Risk - free\ rate$$
$$\sigma = Volatility\ of\ returns\ of\ the\ underlying\ asset$$
$$d_1 = \frac{1}{\sigma\sqrt{T-t}}\left[ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) \times (T-t)\right]$$
$$d_2 = \frac{1}{\sigma\sqrt{T-t}}\left[ln\left(\frac{S}{K}\right) + \left(r - \frac{\sigma^2}{2}\right) \times (T-t)\right]$$
$$d_2 = d_1 - \sigma\sqrt{T-t}$$

© M.M. Rahaman

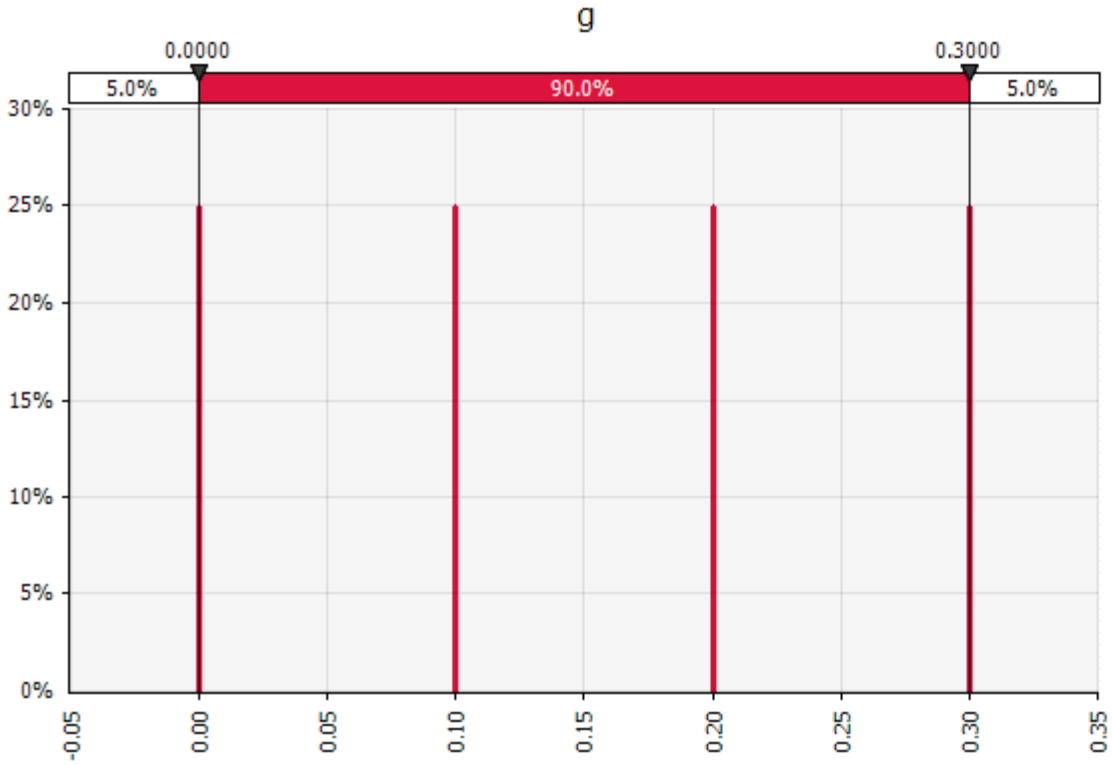
Log-Normal Model of Stock Return Forecast

- On June 30, 2022, Dell Computer stock is sold for $94. An European put option with an exercise price of $80 expiring on November 22, 2022 was selling for $5.25. The current 90-day T-Bill rate is 5.5%. Let us assume that you own a Dell stock on June 30, 2022. What would be your return if you hold the Dell stock till November 22, 2022?

- Refer to the ``LogNormalModel.xls'' file.
- First from the put option find the implied volatility of the Dell stock, that is your σ. You need to familiar with ``Goal Seek'' function in Excel for that.
- Then assume that during the next four months the mean annual return is equally likely to be 0%, 10%, 20%, or 30%.
- If you want to hedge your exposure you can also buy the put option as well.

© M.M. Rahaman

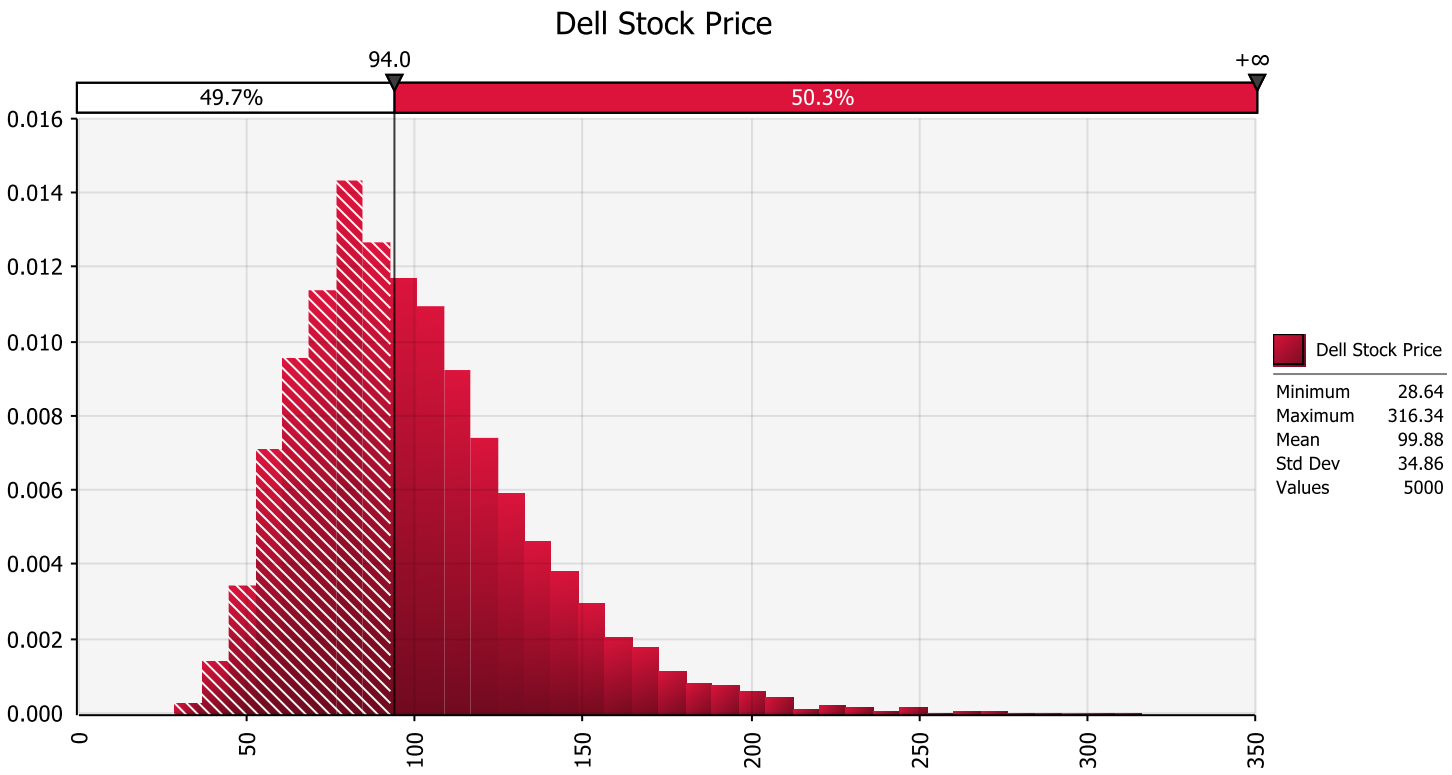
Distribution of μ
Log-Normal Model of Stock Return Forecast



	g
Cell	MODEL!C7
Minimum	0.00000
Maximum	0.30000
Mean	0.15000
90% CI	± 0.00260
Mode	0.00000
Median	0.10000
Std Dev	0.11181
Skewness	0.0000
Kurtosis	1.6398
Values	5000
Errors	0
Filtered	0
Left X	0.0000
Left P	5.0%
Right X	0.3000
Right P	95.0%
Dif. X	0.30000
Dif. P	90.0%
1%	0.00000

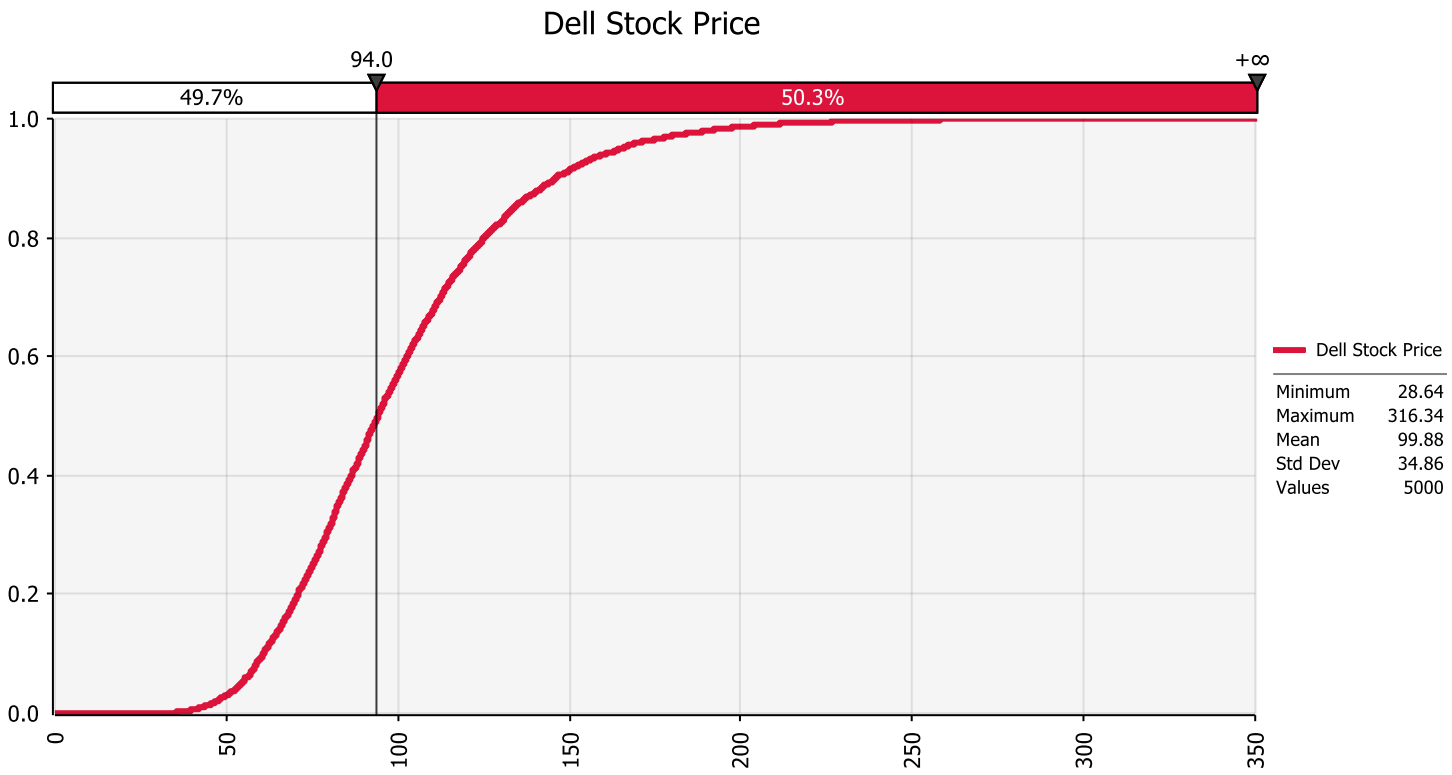
© M.M. Rahaman

Distribution of Dell Stock Price in 4 Months
Log-Normal Model of Stock Return Forecast



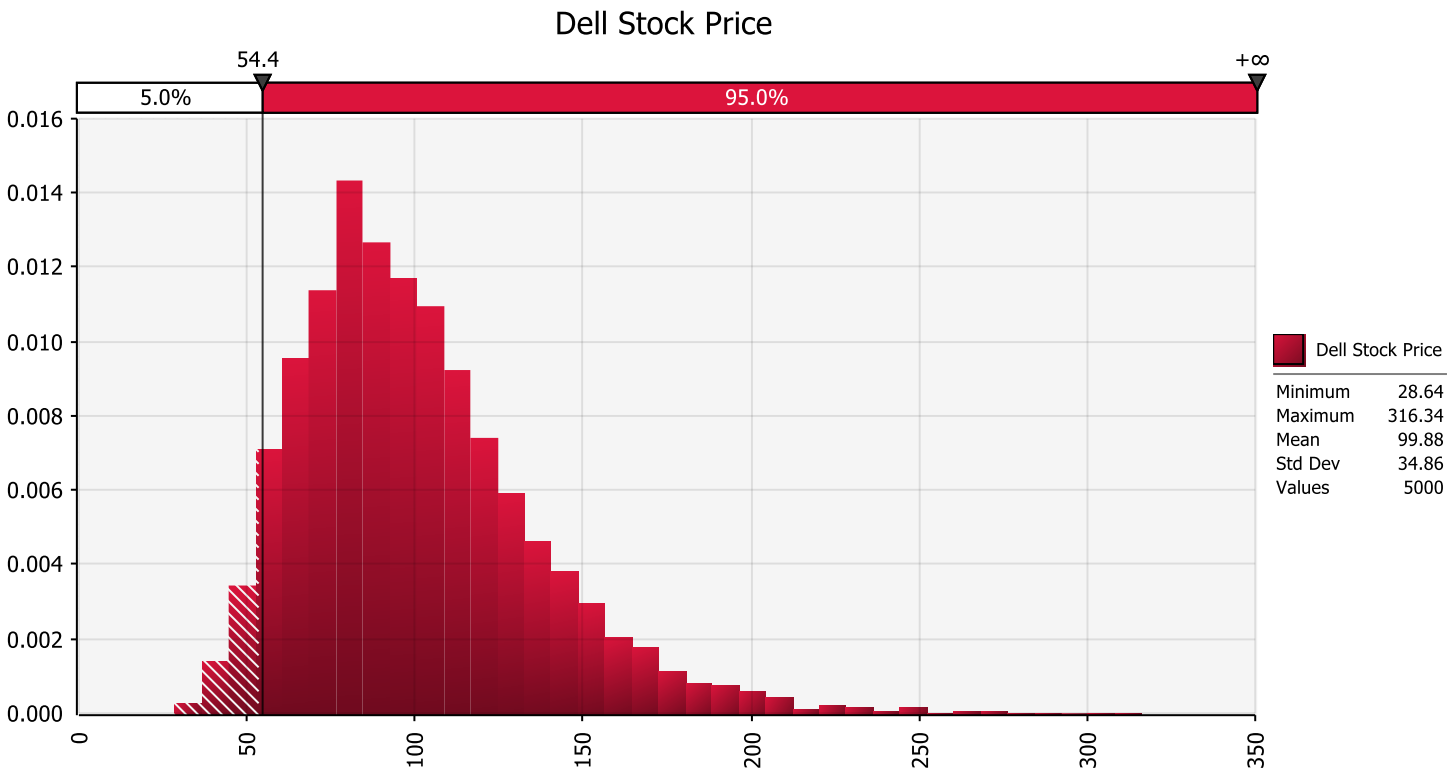
© M.M. Rahaman

PDF of Dell Stock Price in 4 Months
Log-Normal Model of Stock Return Forecast



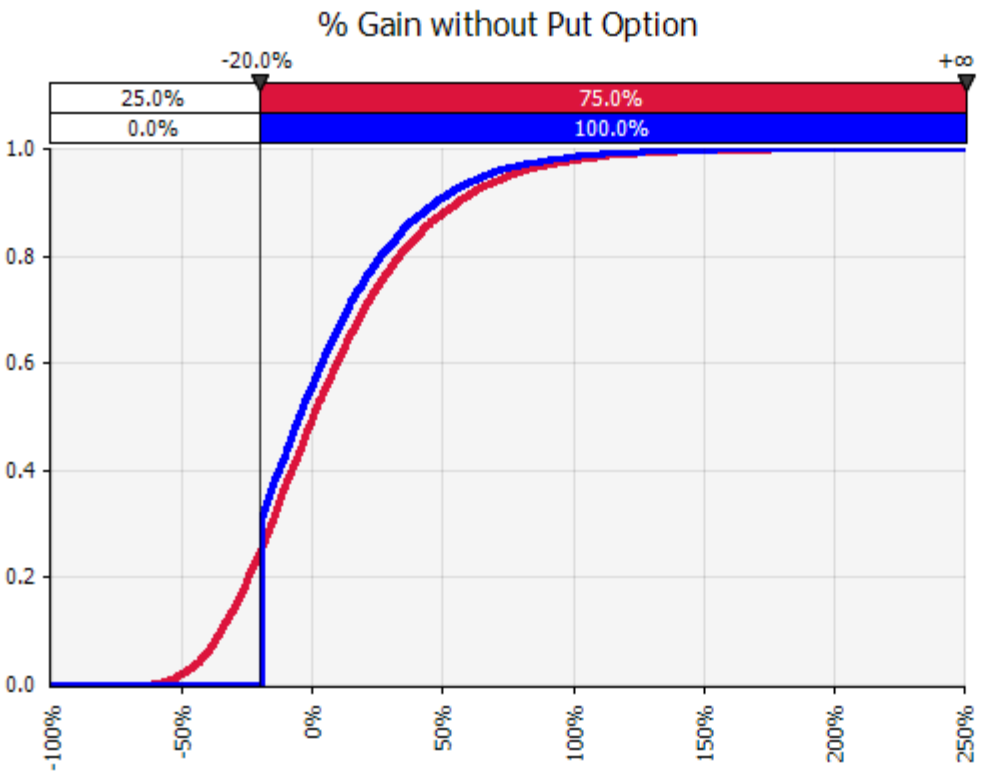
© M.M. Rahaman

5% VaR of Dell Stock Price in 4 Months
Log-Normal Model of Stock Return Forecast



© M.M. Rahaman

Hedging with Put Option
Log-Normal Model of Stock Return Forecast



	% Gain without Put Option	% Gain with Put option hedge
Cell	MODEL!B14	MODEL!B15
Minimum	-69.53%	-19.40%
Maximum	236.53%	218.73%
Mean	6.25%	5.29%
90% CI	± 0.8629%	± 0.7064%
Mode	-5.01%	-19.40%
Median	0.226%	-5.08%
Std Dev	37.09%	30.36%
Skewness	1.0758	1.7275
Kurtosis	4.9608	6.8472
Values	5000	5000
Errors	0	0
Filtered	0	0
Left X	-20.0%	-20.0%
Left P	25.0%	0.0%
Right X	+∞	+∞
Right P	100.0%	100.0%
Dif. X	+∞	+∞
Dif. P	75.0%	100.0%

© M.M. Rahaman

Forecast Stock Return by Bootstrapping

- An alternative approach is to use bootstrapping to model future asset prices.
- Bootstrapping is a statistical technique of resampling.
  - We want to determine the value of a parameter of a population. However, it is typically too expensive or even impossible to measure this directly. So we sample a population, measure a statistic of this sample, and then use this statistic to say something about the corresponding parameter of the population. One of many sampling techniques are known as Bootstrapping.

© M.M. Rahaman

Forecast Stock Return by Bootstrapping

Bootstrap Sampling: We randomly sample from this dataset with replacement to create multiple bootstrap samples, each the same size as the original dataset (100 in this case). For example, one bootstrap sample might look like this:

```
import numpy as np
exam_scores = [60, 95, 72, 94, 76, 61, 85, 83, 81, 86, 63, 95, 66, 80, 82, 88, 84,
               71, 88, 77, 94, 77, 88, 79, 73, 91, 94, 64, 61, 72, 72, 95, 64, 65,
               80, 65, 99, 82, 94, 96, 90, 91, 81, 91, 68, 70, 98, 61, 80, 74, 69,
               69, 74, 81, 73, 75, 64, 96, 89, 91, 96, 79, 71, 67, 86, 89, 90, 65,
               100, 70, 73, 78, 75, 77, 99, 94, 68, 81, 94, 84, 77, 73, 60, 96, 93,
               81, 61, 94, 67, 85, 70, 76, 77, 69, 66, 67, 63, 95, 93, 87]
bootstrap_sample = np.random.choice(exam_scores, size=len(exam_scores), replace=True)
print(bootstrap_sample)
```

```
[ 84  86  71  89  70  71  81  88  89  77  83  74  64  68  71  81  74  80
  91  94  60  69  94  94  96  73  77  69  65  65  70  94  72  95  96  95
  65  73  94  94  61  64  90  63  80  95 100  91  93  81  85  91  84  60
  79  89  73  61  79  90  72  82  78  95  99  83  91  77  73  73  72  91
  76  88  65  99  81  90  85  66  65  88  84  95  77  65  96  81  84  86
  96  94  66  86  95  94  83  96  81  68]
```

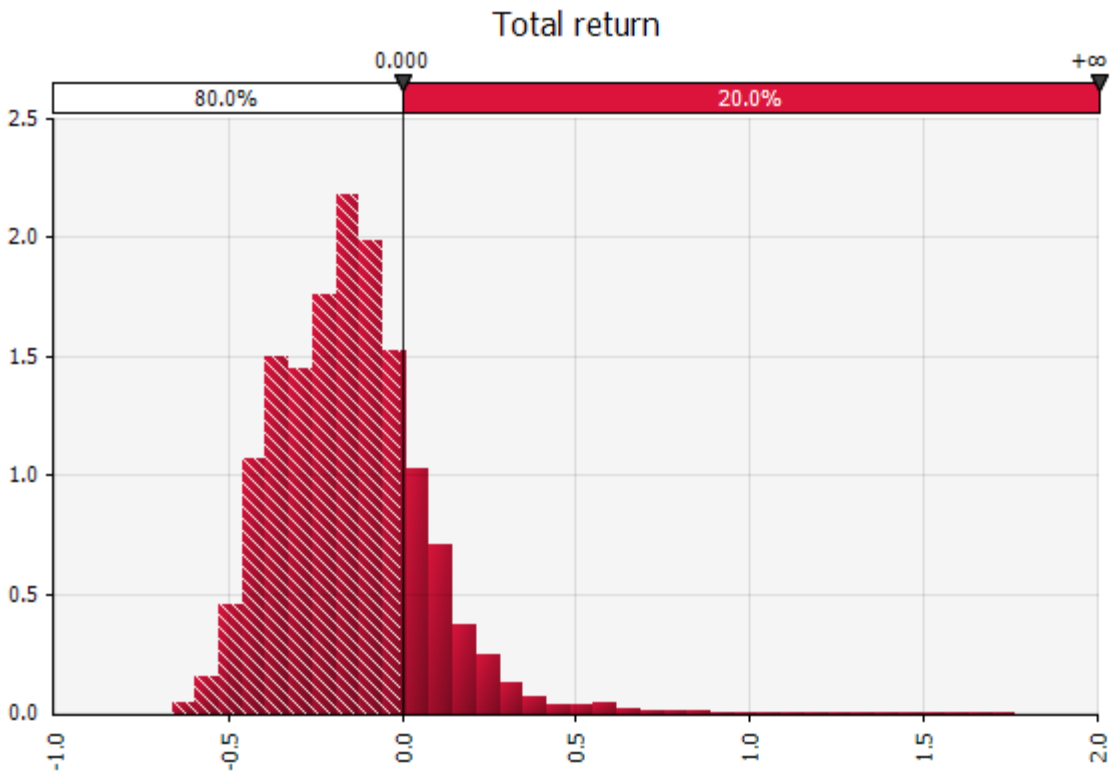

© M.M. Rahaman

Forecast Stock Return by Bootstrapping

- We will use the BootstrapModel.xls file and will simulate six months of stock prices for Microsoft (MSFT), Ford, and Pfizer.
- Lets say today's stock price of MSFT is $79.44, the price of Ford is $43.38, and the price of Pfizer is $50.56.
- Also assume that you want to invest 50% of your money in MSFT, 25% in Ford, and the rest of the 25% in Pfizer.
- Bootstrapping:
  - First, define 62 different scenarios: each date stock price is a scenario.
  - Second, For each of the 126 forecast, randomly draw a scenario from the 62 historical prices with the possibility of resampling.
  - Finally, for each randomly drawn scenario, locate the corresponding return from the historical data and forecast the stock price with that return and previous year stock price.

© M.M. Rahaman

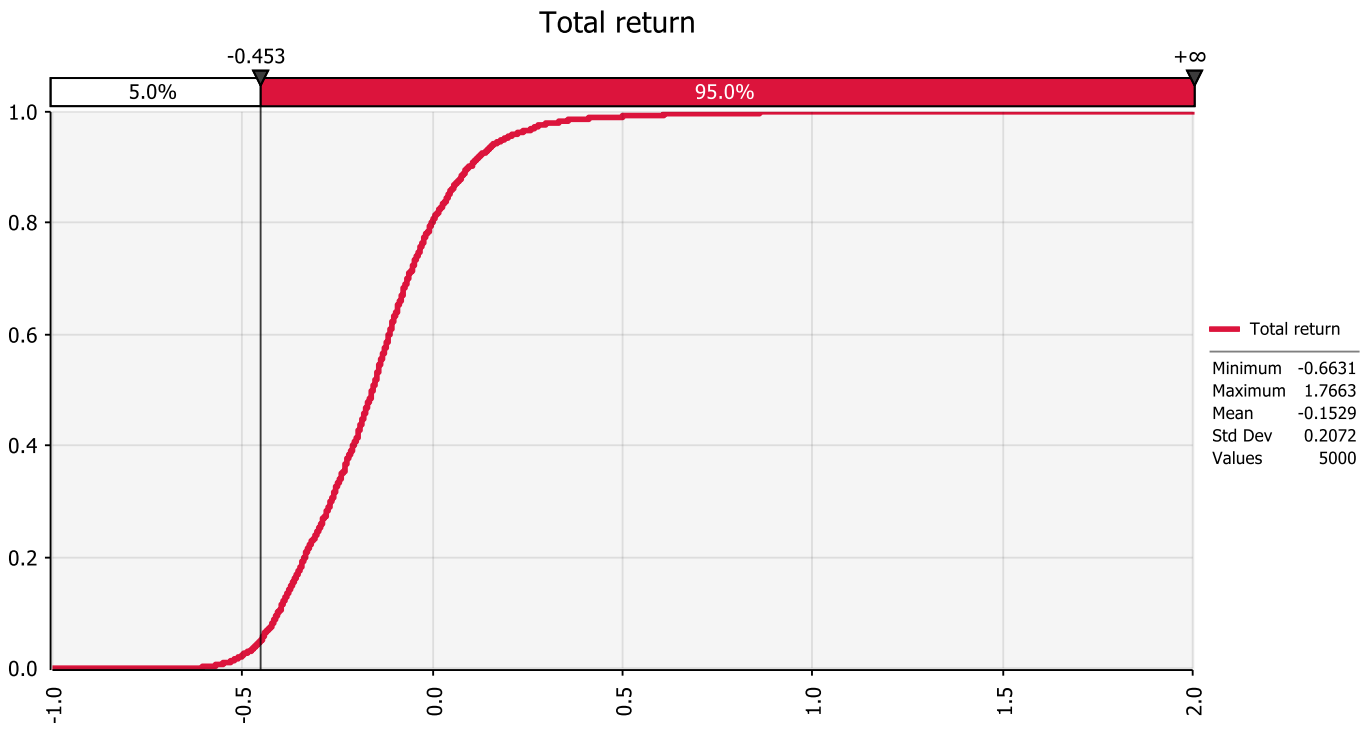
Future Return Distribution
Forecast Stock Return by Bootstrapping



	Total return
Cell	Bootstrap!I135
Minimum	-0.6631
Maximum	1.7663
Mean	-0.1529
90% CI	± 0.00482
Mode	-0.1425
Median	-0.1572
Std Dev	0.2072
Skewness	0.9464
Kurtosis	6.7473
Values	5000
Errors	0
Filtered	0
Left X	0.000
Left P	80.0%
Right X	+∞
Right P	100.0%
Dif. X	+∞
Dif. P	20.0%
1%	-0.5417

© M.M. Rahaman

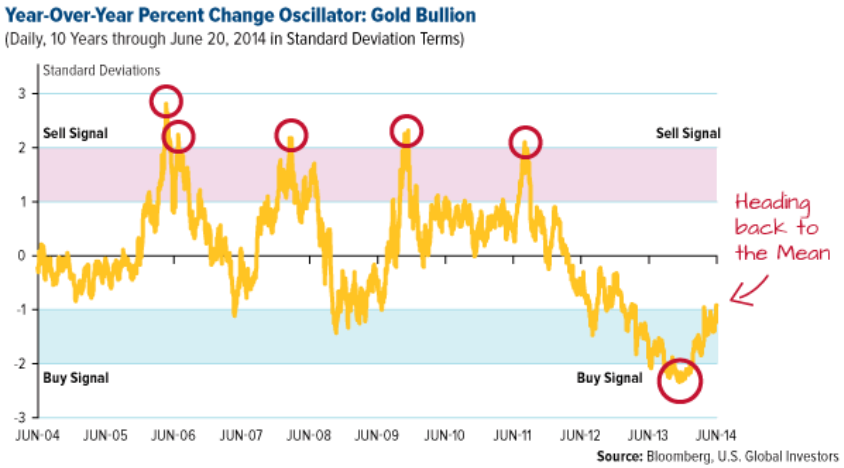
5% VaR
Forecast Stock Return by Bootstrapping



© M.M. Rahaman

Mean-reverting Stock Prices

- We have seen that periods of high stock returns tend to be followed by periods low stock returns while periods of low stock returns tend to be followed by periods of high stock returns.
- This is known as Mean-Reversion and is a popular method to forecast economic variables.



© M.M. Rahaman

Mean-Reverting Stock Prices

Series	Geometric Mean	Arithmetic Mean	Risk Premium	Standard Deviation
Large Stocks	10.23	12.26	8.52	20.82
Small Stocks	12.15	17.80	14.06	40.04
LT Corp Bonds	5.44	5.74	2.00	8.32
LT Govt Bonds	4.85	5.13	1.39	8.00
IT Govt Bonds	5.11	5.24	1.50	5.44
US TBills	3.69	3.74	0.00	3.28
Inflation	3.13	3.23	-0.51	4.68

$$Geometric\ Mean = \left[\left(\prod_{i=1}^{N}(1+R_i)\right)^{1/N} - 1\right] \qquad Arithmetic\ Mean = \frac{1}{N}\sum_{i=1}^{N} R_i$$

Source: Ibbotson Associates/Own Calculations

© M.M. Rahaman

Mean-Reverting Stock Prices

- Please refer to the MeanRevertingModel.xls file
- You have the yearly return on S&P500 index from 1926-2000.
- We are assuming that the mean-reversion cycle occurs every 20 years.
- Therefore, regress the S&P500 return on the 20-year geometric mean to recover the linear forecast parameters as follows:
  - $S\&P500_{Return} = \alpha + \beta \times Year20MeanReturm$
- After you recover the regression parameters, you then use those parameters to forecast the future stock return.
- The details are given in MeanRevertingModel.xls file

© M.M. Rahaman

Building Financial Models

Sobey School of Business
Saint Mary's University

Financial Modelling with Risk & Uncertainty

- Financial Modelling:
  - Limited tools to deal with UNCERTAINTY
  - A lot of tools to deal with RISK
- There is a difference between Systemic Risk and Idiosyncratic Risk
  - Individual variable risk
  - Financial Model risk
- We will spend some time dealing with RISK associated with Financial Modelling

© M.M. Rahaman

Financial Modelling with Risk & Uncertainty

- Financial Modelling:
  - Pick your valuation techniques:
    - Discounted Cash Flow Method: You need to forecast your cash flow and the discount rate.
    - Relative Valuation Method: You need to pick your valuation multiple and comparable firms.
    - Residual Income Valuation Model: You need a way to forecast you residual income and a discount rate
    - Contingent Claim Valuation Model: You need to well-versed in valuating financial derivatives to implement this type of model.
  - Whatever valuation technique you decide to use (most popular one is the DCF), always start with a benchmark/static model such as a multi-stage DCF model.
  - Progressively build you model with underlying assumptions about your inputs and outputs.
  - Perform risk analysis of your model to examine the ROBUSTNESS of your Financial Model. You need to have a strong sense of distributional structure of your risk and uncertainty. Please see the MFIN6690_ProbDistribution lecture.

© M.M. Rahaman

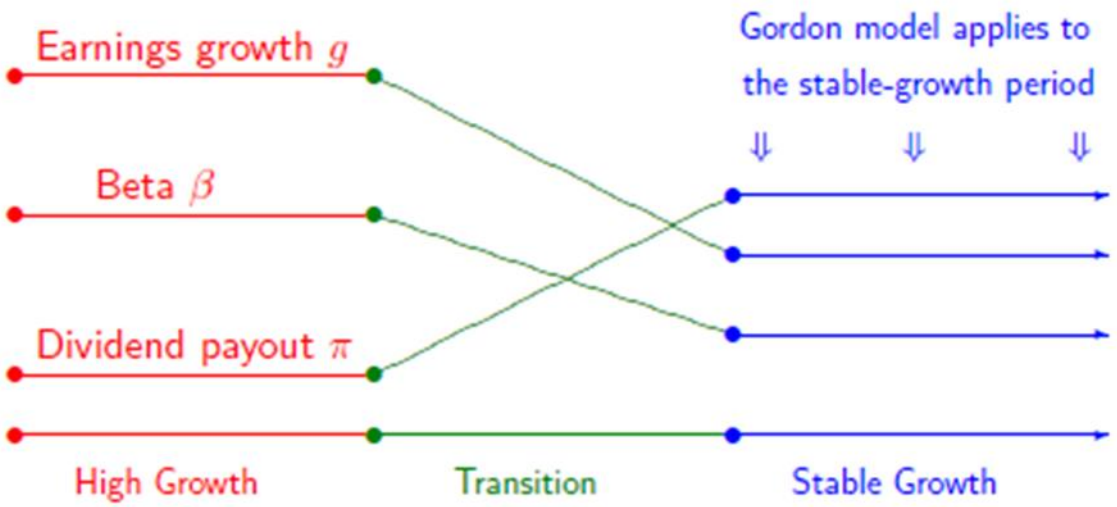
Building Blocs of A Financial Model

- Building Blocs of the Benchmark Financial Model:
  - Start some KEY information. **Know the difference between DATA and INFORMATION.**
  - Identify KEY inputs of your financial model.
  - Make reasonable ASSUMPTIONS about the DYNAMIC PATH of the key inputs in your financial model.
  - Based on your assumptions **FORECAST** the future paths of your key inputs.
  - Make sure that your model is INERNALLY CONSISTENT.
  - Calculate the OUTPUT of your model.
  - Perform comparative-static analysis – Sensitivity, Scenario, and Smilation

© M.M. Rahaman

Benchmark Financial Model – An Example

- A popular extension of the Gordon model is the three-stage DDM.
- The three-stage DDM allows for an initial period of high growth, a transitional period when growth declines, and a final stable-growth phase.



© M.M. Rahaman

Benchmark Financial Model – An Example

- Estimating the equity value of Home Depot
- Information about Home Depot:
  - Current earnings and dividends: $E_0 = 1.50$, $D_0 = 0.18$
  - CAPM risk premium and risk-free rate are equal to **6% and 5%**, respectively.
- Assumptions for the inputs:
  - The high-growth period (of five years): $g = 36\%$, $\pi = D_0/E_0 = 12\%$, and $\beta = 1.6$
  - The transitional period (of five years)
  - Earnings growth rate will decline from **36%** in year 5 to **6%** in year 11 in linear increments.
  - Over the same time interval, payout ratio will increase from **12% to 60%**, and beta will decline from **1.6 to 1.0**.
  - The stable-growth period: $g = 6\%$, $\pi = 60\%$, and $\beta = 1.0$

© M.M. Rahaman

Benchmark Financial Model – An Example

- Calculating the Present Values of Dividends:

Year	g	EPS	π	DPS	β	r	PV of DPS
1	36%	2.0400	12%	0.2448	1.6	0.146	0.214
2	36%	2.7744	12%	0.3329	1.6	0.146	0.254
3	36%	3.7732	12%	0.4528	1.6	0.146	0.301
4	36%	5.1315	12%	0.6158	1.6	0.146	0.357
5	36%	6.9789	12%	0.8375	1.6	0.146	0.424
6	31%	9.1423	20%	1.8285	1.5	0.140	0.811
7	26%	11.5193	28%	3.2254	1.4	0.134	1.262
8	21%	13.9384	36%	5.0178	1.3	0.128	1.741
9	16%	16.1685	44%	7.1142	1.2	0.122	2.200
10	11%	17.9471	52%	9.3325	1.1	0.116	2.586

© M.M. Rahaman

Benchmark Financial Model – An Example

For example, for year 7, the earnings is $E_7 = 1.5(1.36)^5(1.31)(1.26) = 11.5193$.

With the payout ratio of 28%, the PV of the dividend for year 7 is

$$\text{PV of } D_7 = \frac{0.28 \times 11.5193}{1.146^5 \times 1.140 \times 1.134} = 1.262.$$

Applying the Gordon model, the terminal value at the end of year 10 is

$$\text{Terminal Value} = \frac{17.9471 \times 1.06 \times 0.60}{0.11 - 0.06} = 228.287.$$

The PV of dividends in both the high-growth and the transition periods =\$10.150.

The PV of terminal value at the end of the transition =\$63.252.

The intrinsic value of the stock = \$73.402.

© M.M. Rahaman

Benchmark Financial Model – An Example

- Probabilistic approaches:
  - Sensitivity analysis
  - Scenario analysis
  - Monte Carlo simulation
  - Decision tree analysis

© M.M. Rahaman

Sensitivity Analysis

- How sensitive your output is to the changes in an input of your model?
  - Mathematically: $\frac{\text{\% change in output}}{\text{\% change in input}}$
  - Assume "Y" is output and "X" is input, then the sensitivity is calculated as: $\frac{\Delta Y}{\Delta X} \times \frac{X}{Y}$

© M.M. Rahaman

Steps in Sensitivity Analysis

- **Step 1:** Determine the factor around which the sensitivity analysis will be built. Typically, focus on one, two, or (at most) three critical factors/inputs in the model.
- **Step 2:** The second component is determining the number of variations to analyze for each factor/input used in the sensitivity analysis.
- **Step 3:** The third component is the estimation of the output of the model for each possible variations in the critical factors/inputs.
- **Step 4:** The final component is the assignment of probabilities to each possible outputs generated from the variations in the critical factors/inputs.

© M.M. Rahaman

Sensitivity Analysis: Excel Examples

- These are some of the Excel-based sensitivity analyses that you need to understand to get sense of how such probabilistic analysis is being conducted:
  - See the FinModelling_3SDDM.xlsx
  - See the NewProductLaunch_SensitivityAnalysis.xlsx
  - See the Loan_SensitivityAnalysis.xlsx

© M.M. Rahaman

Scenario Analysis

- Classify the range of sensitivity analysis into different scenarios.
- For each scenario, we vary the critical inputs/factors in the model to generate scenario-based output.
  - Examples: Best-case scenario, Worst-case scenario, Most-likely scenario

- Steps in Scenario Analysis:
  - **Step 1:** Identify how many scenarios to analyse.
  - **Step 2:** For each scenario, identify the critical inputs/factors to vary to generate the scenario based output.
  - **Step 3:** Do sensitivity analysis by varying the critical inputs/factors for a given scenario and generate the output from the model.
  - **Step 4:** The final component is the assignment of probabilities to each possible outputs generated from the pre-defined scenarios.

© M.M. Rahaman

Scenario Analysis: Examples

- These are some of the Excel-based scenario analyses that you need to understand to get sense of how such probabilistic analysis is being conducted:
  - See the FinModelling_3SDDM.xlsx
  - See the NewProductLaunch_ScenarioAnalysis.xlsx

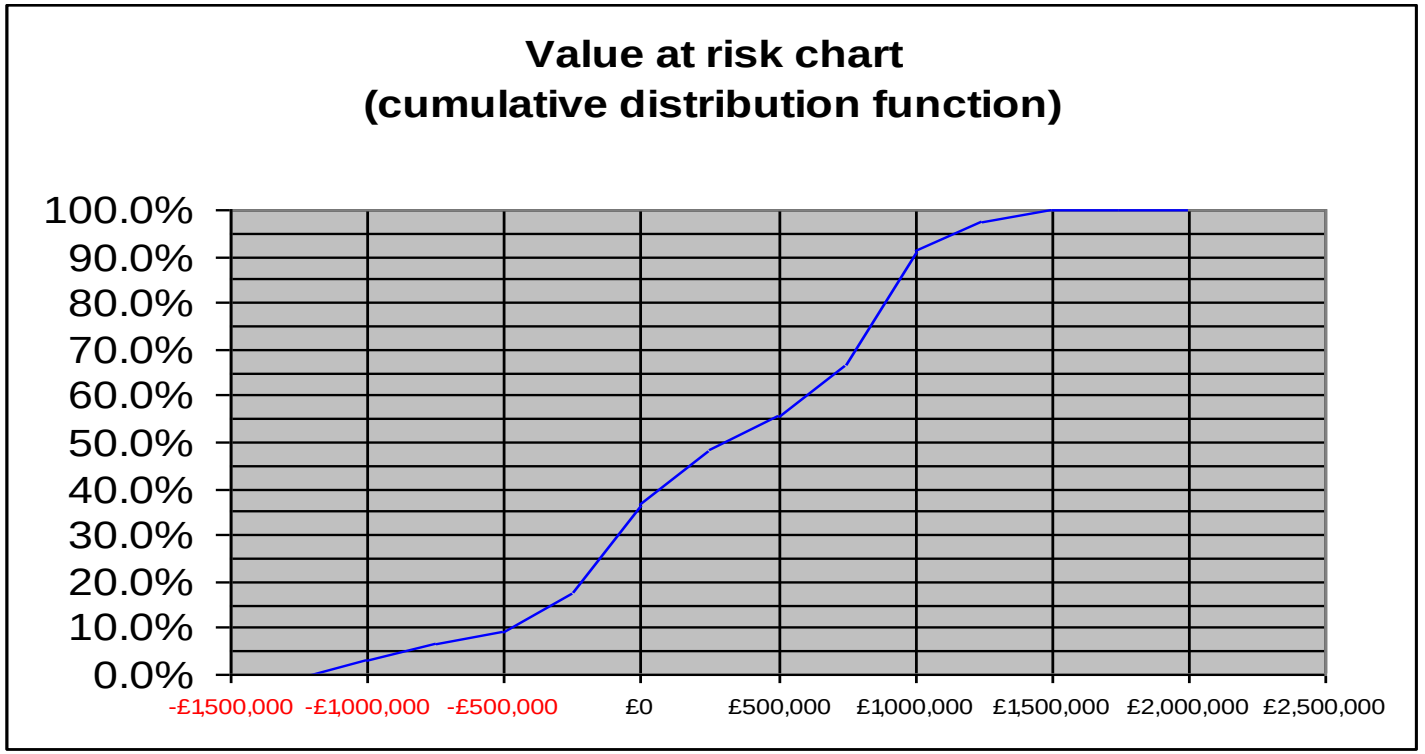
© M.M. Rahaman

Monte Carlo Simulation

- **Sensitivity and scenario analysis:**
  - Allows for simple graphical representation only if there is only ONE piece of uncertain input data (e.g. NPV as function of initial demand)
  - If there are more than one uncertain inputs, it is not possible to visualize uncertainty in a graphical manner.
  - Thus, difficult to get an idea how OUTPUT changes if you have many INPUTS to play with.
- **Monte Carlo Simulation (MCS)** is automated what-if analysis that takes probabilities into account
  - Allows for simple graphical representation (histogram, cumulative distribution function of profit) even if there are MULTIPLE uncertain inputs.

© M.M. Rahaman

Monte Carlo Simulation: Distribution & Value-at-Risk (VaR)



10% VAR is roughly £500,000

5% VAR is roughly £800,000

© M.M. Rahaman

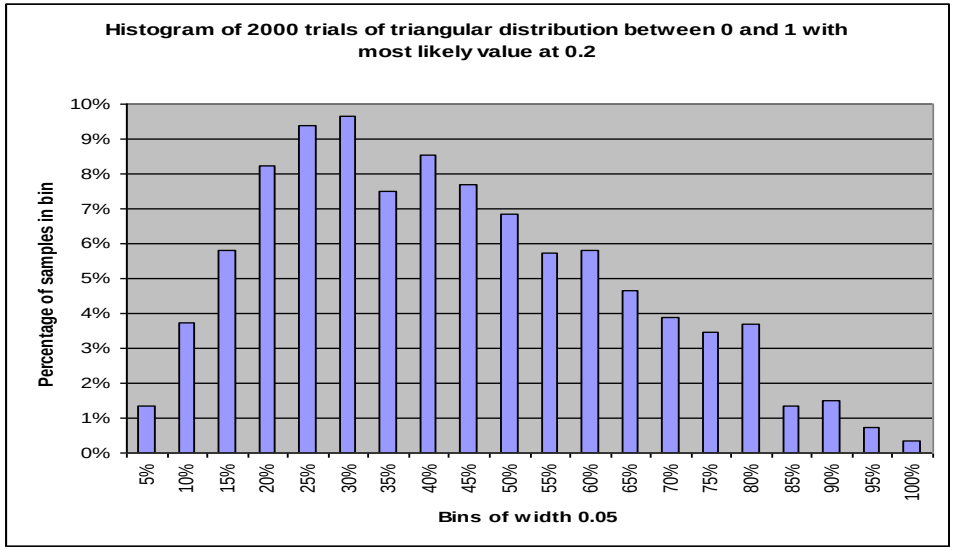
How Does MCS work?

- Start from logically correct spreadsheet model
  - Some cells in your spreadsheet are filled with uncertain numbers (demands, raw material prices, discount rate etc.)
- Determine sensible distributions for these uncertain numbers. This is a bit tricky and requires experience and judgement.
  - Base distributional assumptions on **historic data** or **subjective judgement**.
- Ask your computer to sample many (e.g. 1000) scenarios randomly from these distributions.
- Record the value of the baseline (e.g. NPV) for each generated scenario.
- Display the *distribution* of the baseline (over all samples)
  - Teach your clients to read cumulative distribution functions or histograms.
- You avoid the flaw of averages in a Monte Carlo Simulation.

© M.M. Rahaman

Critical Inputs in a MCS Model

- A logically correct model such as Profit calculation, Stock price calculation, NPV calculation, and portfolio return calculation.
- A set of uncertain variables in the model such as Cash-flow, Discount rate, Demand, Production cost.
- An approximate description of the distribution of the uncertainties. Very Critical.
  - I personally like to ask for a triangular distribution



© M.M. Rahaman

Monte Carlo Simulation: A Bit About Distribution

- Normal distribution arises naturally, due to the central limit theorem.
  - Mean and standard deviation need to be estimated.
- Discrete distributions are sensible if one has only a few scenarios to consider.
  - However, use of subjective probabilities is problematic and requires careful sensitivity analysis.
- Empirical distributions are sensible if one has plenty of data and time horizon is short.
  - To sample from an empirical distribution of m data points, you choose a number between 1 and m and sample the mth data point.
- The calculation of an expected value is basically an integration over the distribution.
- When dimension is high, simple numerical integration methods are likely to be impractical.
- Monte Carlo methods for computing integrals can deliver moderately accurate solutions.
- The motivation for the use of Monte Carlo integration is the **Law of Large Numbers**.

© M.M. Rahaman

Why MCS Works: The Law of Large Number

- If X_i is a collection of i.i.d. random numbers with density *q(x)* and support [0,1], then:

$$\lim_{N\to\infty}\frac{1}{N}\sum_{i=1}^{N}X_i=\int_0^1 xq(x)dx$$

- Furthermore:

$$Var(N^{-1}\sum_{i=1}^{N}X_i)=\frac{\sigma_x^2}{N}$$

- If the variance of X is unknown a priori, an unbiased estimator is:

$$\hat{\sigma}_x^2\equiv(N-1)^{-1}\sum_{i=1}^{N}(X_i-\overline{X})^2$$

© M.M. Rahaman

Why MCS Works: The Central Limit Theorem (CLT)

- Let $\boldsymbol{X_1, X_2, \ldots \ldots, X_n}$ be i.i.d random variables with the same mean $\boldsymbol{\mu}$ and the same standard deviation $\boldsymbol{\sigma}$. Furthermore, let $\boldsymbol{Y_n = X_1 + X_2 + \cdots + X_n}$.

- Then the new random number $\boldsymbol{Z_n = \frac{Y_n - n\mu}{\frac{\sigma}{\sqrt{n}}}}$ is asymptotically normally distributed with mean **0** and variance **1**.

© M.M. Rahaman

MCS: How Many Replications are Necessary?

- There is no strict answer to this question.
  - Trade-off between speed and accuracy
- Important rule of thumb: use more replications than you think necessary (several thousands).
- An accuracy check: Take sample mean and standard error of the measurements.
  - Sample mean is normally distributed with expectation being the true (unknown) mean of the population and standard deviation being the standard error = standard deviation of the sample / SQRT(sample size).
  - Mean is normally distributed, hence 95% confidence interval is mean +/- 2* Standard Error.

© M.M. Rahaman

MCS: Be Careful about Statistical Dependence

- Your model is likely to contain several sources of uncertainty which are statistically dependent:
  - Demand, costs, etc. in periods 1,2,3,...N.
  - Demand and price in a single period.

- It is crucially important to take statistical dependence into account.
  - See spreadsheet example of Monte Carlo Simulation.

- Statistical dependence can be taken into account by building a model of the uncertainties
  - Regression model Y=f(X) + random effect.
  - Dynamic model: Demand in year n+1=demand in year n + random effect (lattice model).

© M.M. Rahaman

Trouble with MCS: Incorporating decision rule/Managerial flexibility

- MCS is great for analysing the effect of uncertainty.
- MCS is not that well suited for analysing the effect of flexibility.
  - Value of flexibility is linked to optimising one's actions.
- MCS is not well suited for optimization.
  - Not surprising since the essence of MCS is that it does not give you a single value that you can optimise but a whole distribution (which you can't "optimise" in the same way).
- **Decision Tree Analysis** is better suited for optimisation (however based on single numbers, e.g., expectations, at each optimization node). Decision trees allow us to visualize the interplay between uncertainty and flexibility.

© M.M. Rahaman

Decision Tree Analysis: Decision-event sequences

- A path through the tree is a decision-event sequence.
- There are (generally) two types of nodes (forks)
  - Decision nodes: You make a decision here.
  - Event nodes (chance nodes): Some event happens here.
- There are quantitative Information on arcs and nodes.

© M.M. Rahaman

Decision Tree Analysis: Valuing the Tree

- Each node represents:
  - a point in time.
  - a possible state of the system at that point in time.
- Valuation approach: each node has a value
  - Leafs of the tree have values: sum of discounted cash flows along the path from the origin to the leaf.
- Backwards induction:
  - Break-down the whole decision tree into sub-tress.
  - Value of a sub-tree is the expected value from the payoff of the sub-tree.
  - Push the values node-by-node backwards through the tree to the origin.
  - Collapse the tree through successive elimination of forks.

© M.M. Rahaman

Decision Tree Analysis: Valuing the Tree

- The following Excel-based example will help you to understand the basics of decision tree development, valuation, and analysis:
  - See example: BasicDecisionTree_Example.xlsx
  - See example: DecisionTreeAnalysis_AmericanPutOption.xlxs

© M.M. Rahaman